

Learning from the Rare



Talk at IIM Bangalore

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Outline



❖ Introduction

❖ Motivating Examples

❖ Hellinger Net (Proposed Solution)

❖ Application and Deployment

❖ Discussions



Introduction & Motivation

- Machine learning systems are used in the fields of science, technology, and medicine, mainly for their accuracy and ability to deal with complex data structures.
- ML systems largely depend on the quality of training data, ideally a training set is expected to adhere to some regularity conditions.
- I develop theoretically-principled machine learning systems that are scalable, robust, accurate, and easily interpretable to address challenges of data irregularities in real-world data.
- I study theoretical (SLT and asymptotic results in the form of so-called error bounds), computational, and applied aspects of our developed frameworks.

ForeML Lab

“The scope of statistics extends over the whole range of the natural and social sciences, engineering and technology, medicine, management and economic affairs” - Professor PC Mahalanobis.



Macroeconomic
Forecasting



Multivariate Time
Series Analysis



Financial ML



Manufacturing Process
Efficiency Improvement



Imbalanced Software
Defect Prediction



Analysis of Networks

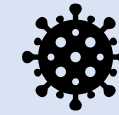
Engineering

Economics

STAT+ML

Healthcare

Geoscience



Epidemic Forecasting



Reinforcement Learning
for Mobile Health



Digital Intervention
Trial Design



Generative AI for
Urban Science



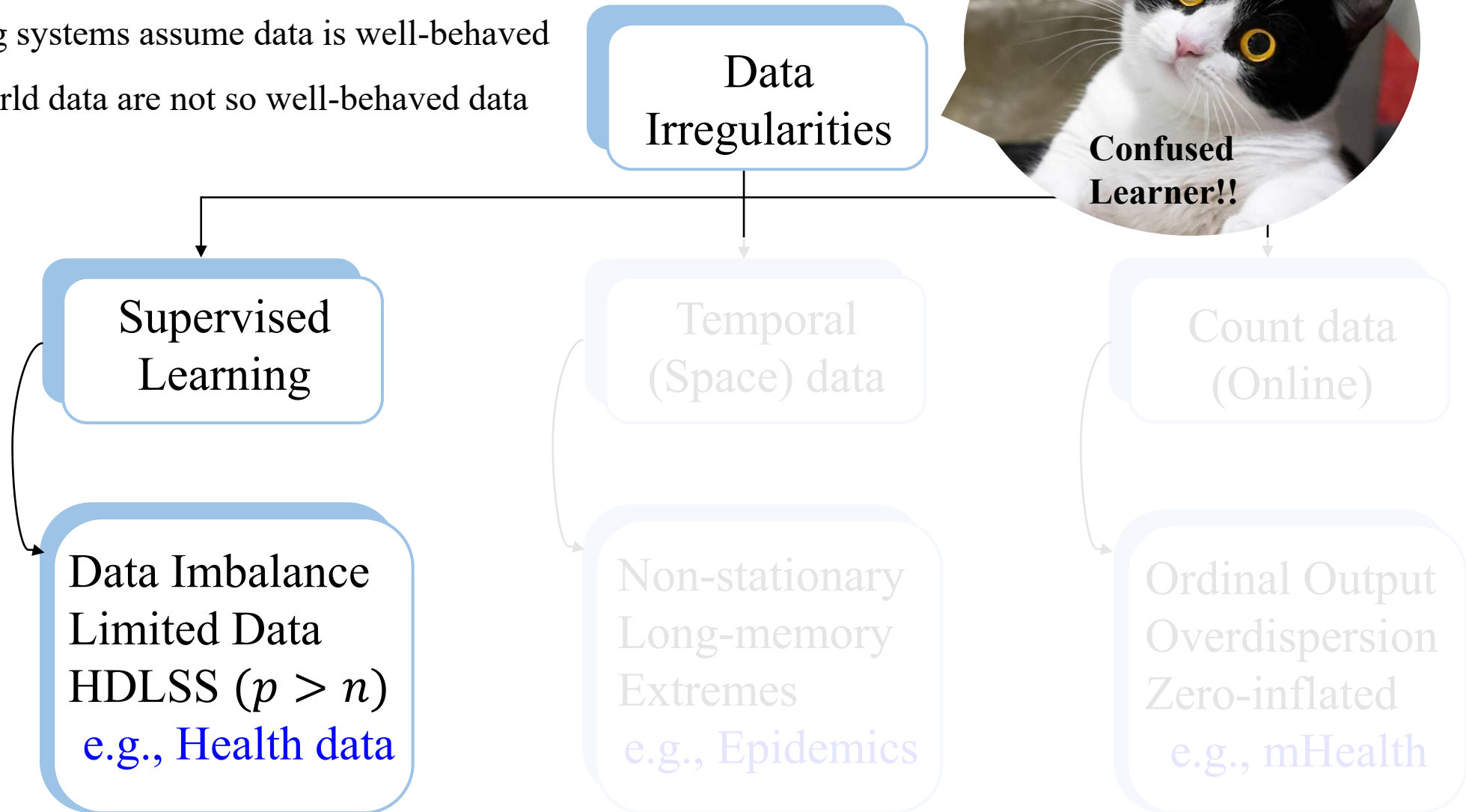
Modeling Air
Quality Data



Understanding
Climate Change

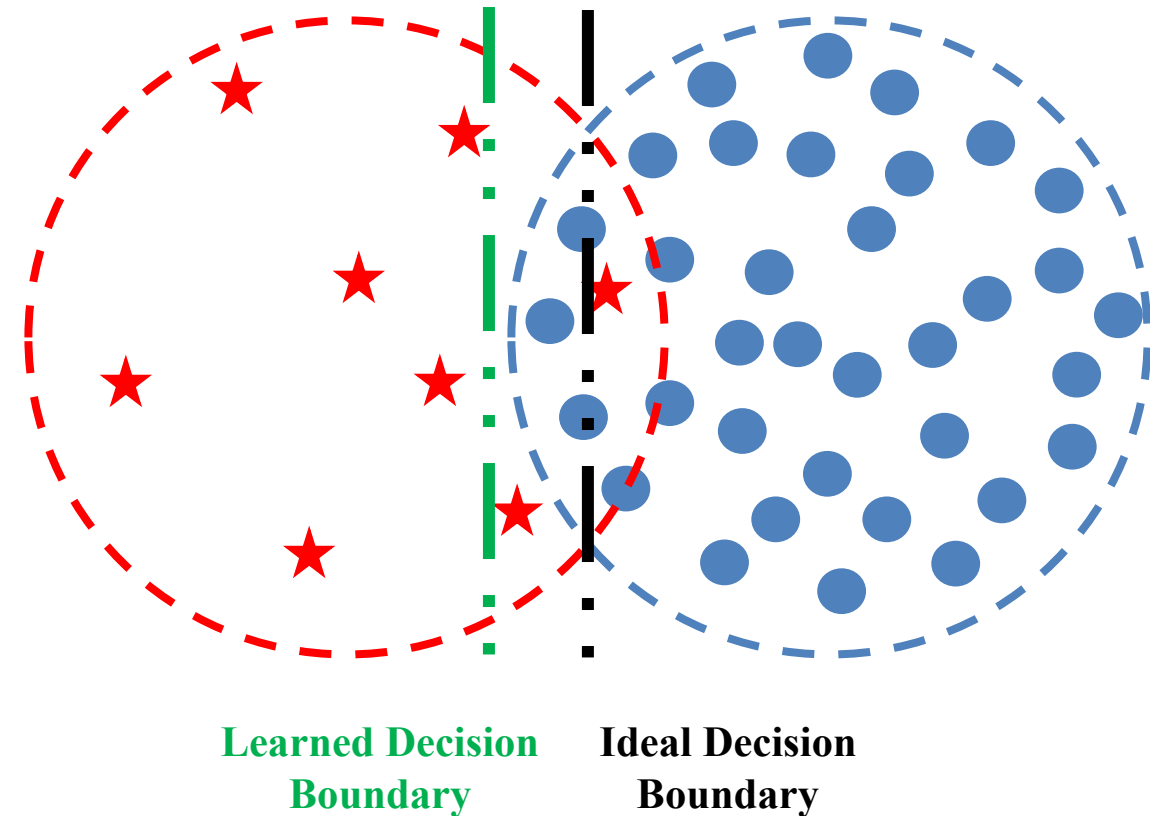
Data Irregularities

- Learning systems assume data is well-behaved
- Real-world data are not so well-behaved data



Imbalanced Classification Problem

- Many real-world data sets are usually skewed, in that many cases belong to a larger class (**majority class**) and fewer cases belong to a smaller yet usually more exciting class (**minority class**).
- This leads to **class imbalance problem**.
- **Example:** Detecting rare events: uncommon diseases data having only a few sick persons and a lot of healthy persons; Cyclone Prediction from satellite imagery.
- Traditional classifiers usually aim to optimize the overall accuracy without considering the **relative distribution of each class**.



Imbalanced data with Limited sample size

Imbalance Ratio (IR): $1,000,000 : 10,000 = 100 : 1$?

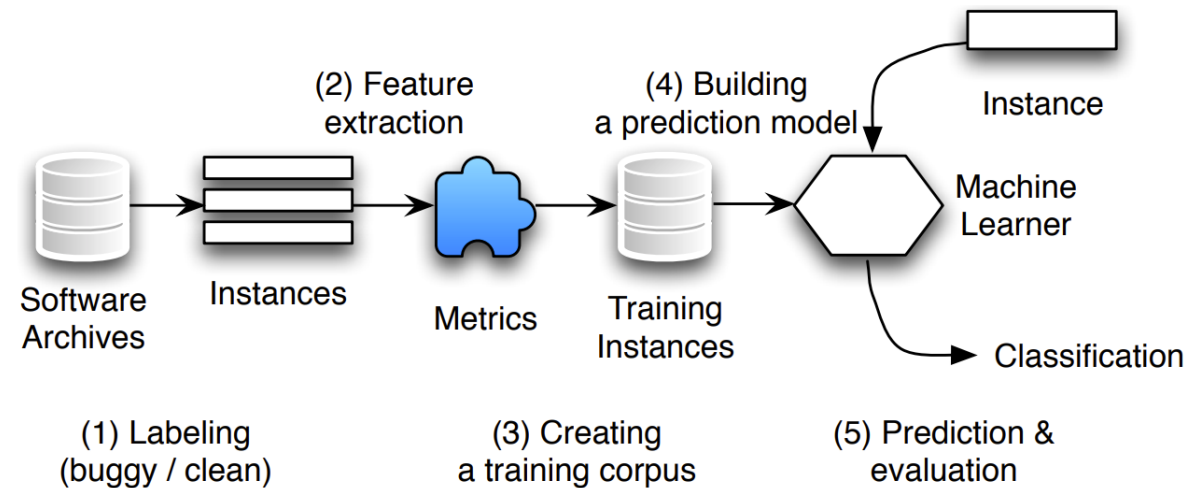
- The minority class may be outnumbered, but not necessarily rare
- Therefore, they can be accurately learned with little disturbance

Imbalance Ratio (IR): $2,000 : 20 = 100 : 1$?

- Highly imbalanced data with limited sample size
 - Alzheimer's disease detection, Gene expression, Fertility detection
- Challenges with limited sample size imbalanced data for learning systems:
 1. Failure of generalizing inductive rules
 2. Difficulty in forming “good” decision boundary
 3. Risk of overfitting

Motivating Example: SDP Dataset

- **Software Defect Prediction** (SDP) is the deployment of learning systems for identifying defective software modules or components using software features.
- Example is **ISRO/NASA SDP datasets**, where Software metrics are defined by McCabe and Halstead describing the attributes of a module to find the defect.
- Binary classification problem in which software module is classified as defect-prone or non-defect-prone.
- **Benefits:** Cost effective, producing quality software, manpower allocation.



NASA SDP Datasets



Data set	Classes	Objects	Attributes	Defect %
CM1	2	327	37	12.8
KC3	2	194	39	18.6
MC1	2	1988	38	2.3
MC2	2	125	39	35.2
MW1	2	253	37	10.7
PC1	2	705	37	8.7
PC2	2	745	36	2.1
PC3	2	1077	37	12.4
PC4	2	1287	37	13.8
PC5	2	1711	38	27.5

Motivating Example: Thyroid Gland Dataset

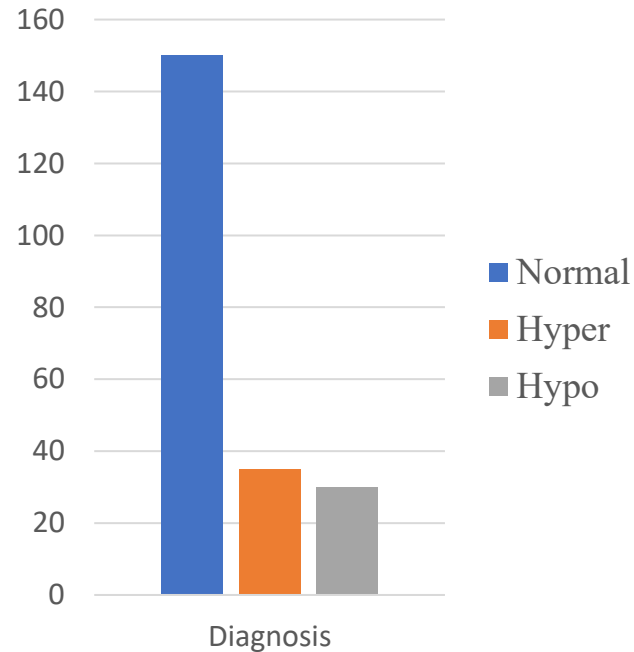
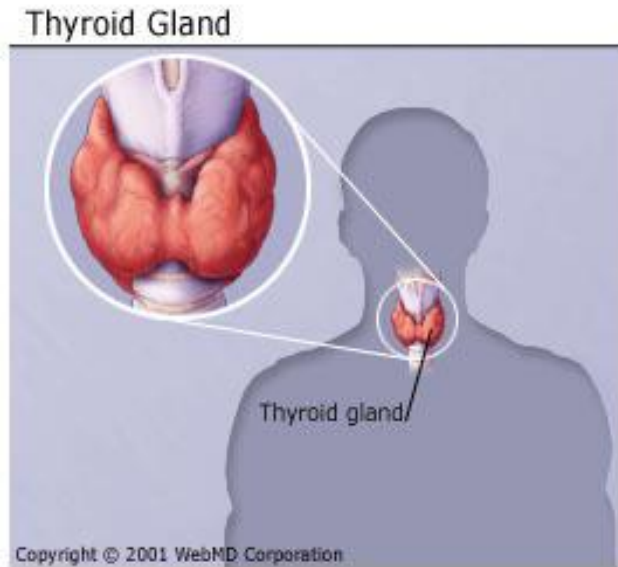


Fig: Output class distribution

- Data on five laboratory tests administered to a sample of 215 patients.
- Aim is to predict whether a patient's thyroid can be classified as:
 - **euthyroidism** (normal thyroid gland function),
 - **hypothyroidism** (underactive thyroid not producing enough thyroid hormone),
 - **hyperthyroidism** (overactive thyroid producing excessive amounts of the free thyroid hormones T3 and/or thyroxine T4).

Thyroid Gland Data: Features

Diagnosis of thyroid operation was based on the following medical records (numerical values):

- RT3U - T3-resin uptake test (percentage).
- T4 - Total Serum thyroxin as measured by the isotopic displacement method.
- T3 - Total serum triiodothyronine as measured by radioimmuno assay.
- TSH - Basal thyroid-stimulating hormone (TSH) as measured by radioimmuno assay.
- DTSH - Maximal absolute difference of TSH value after injection of 200 micro grams of thyrotropin-releasing hormone as compared to the basal value.

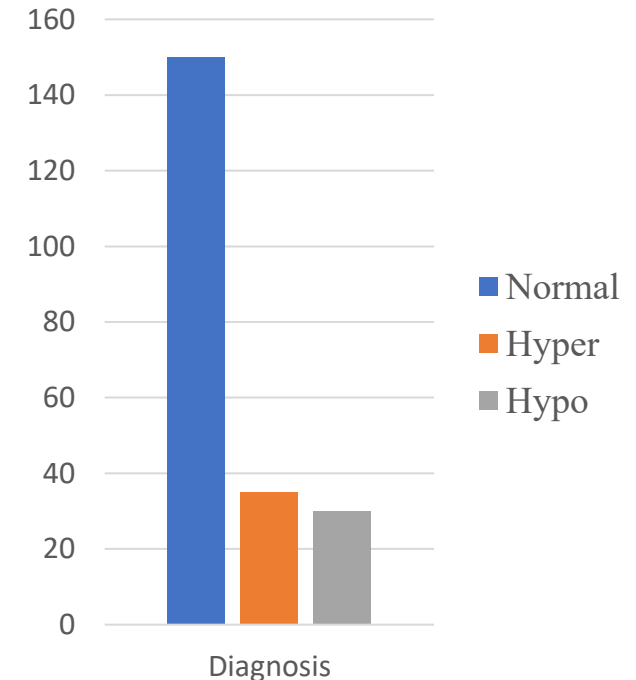
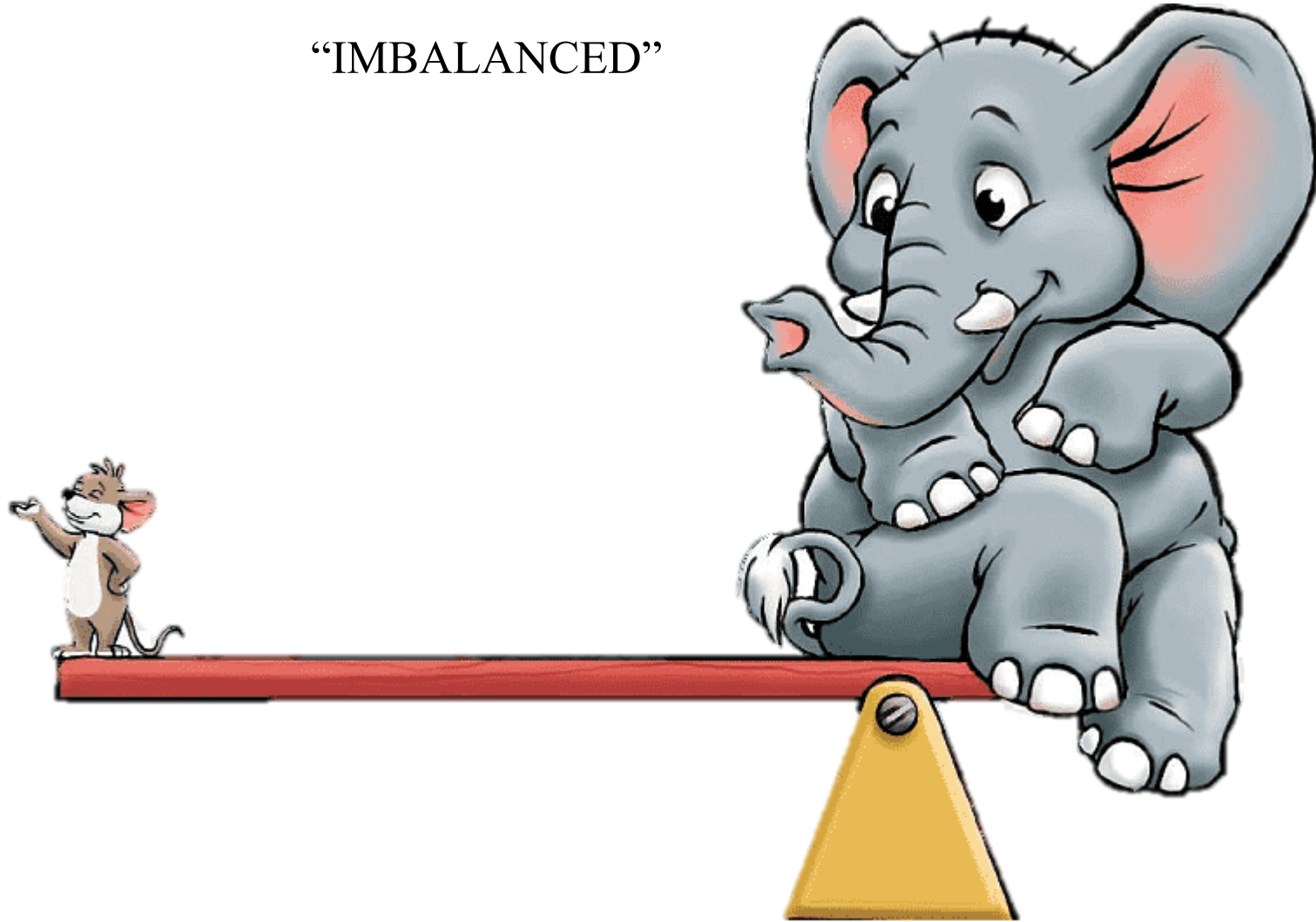


Fig: Output class distribution

Class Imbalance



“IMBALANCED”



Effect of Class Imbalance on Neural Networks

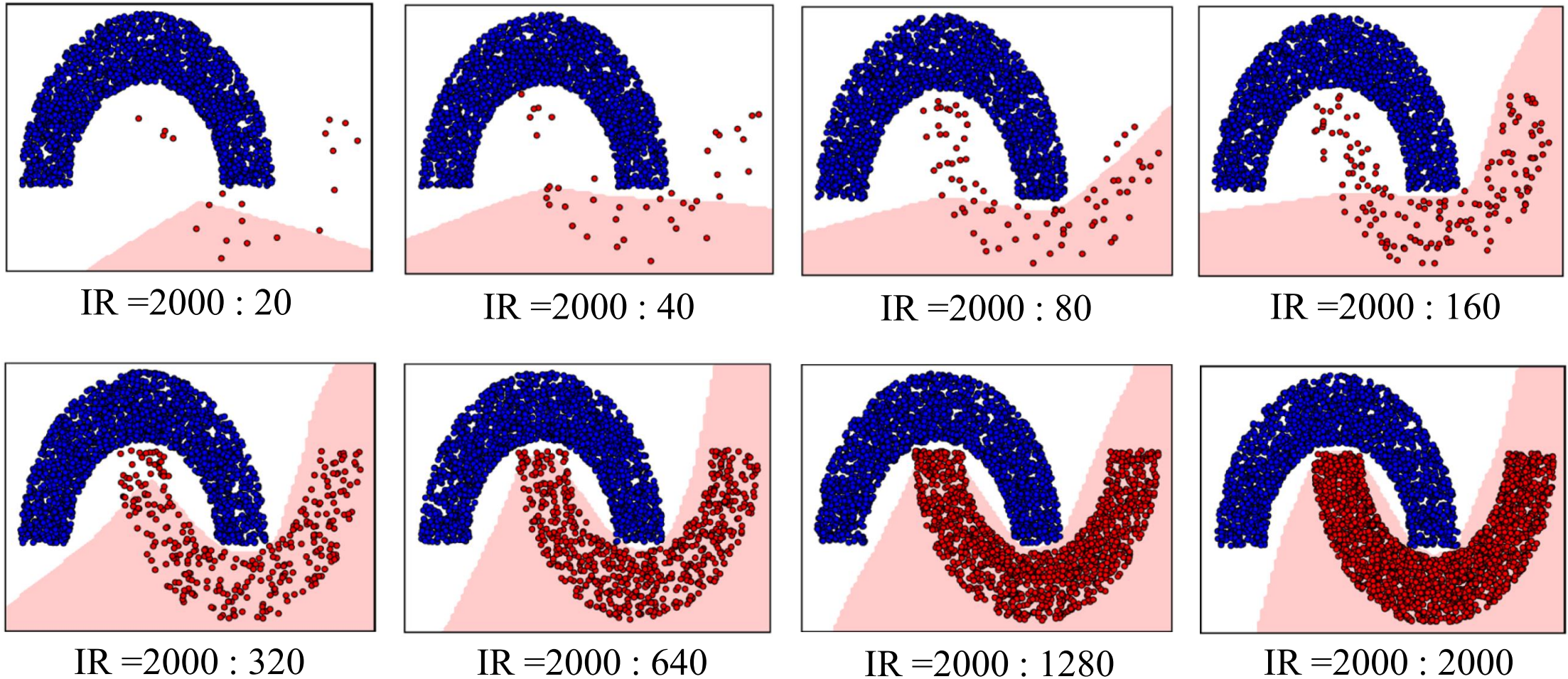


Fig: MLP Classifier (2 HL) on Two moon dataset with interleaving classes.

Effect of Class Imbalance on Decision Tree

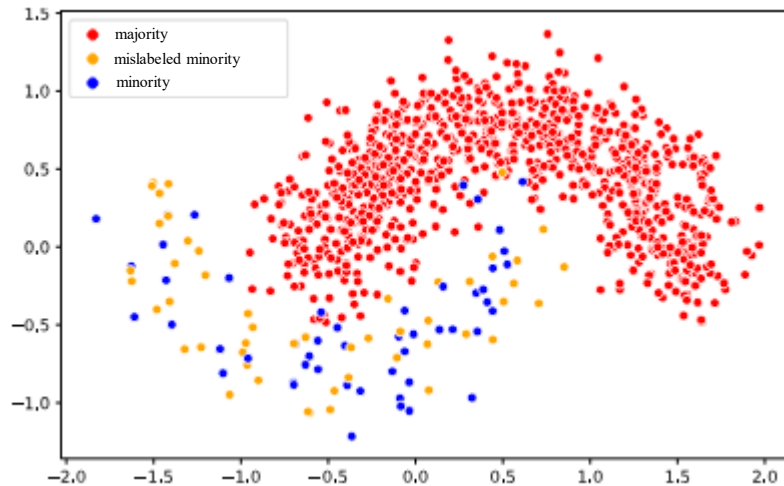


Fig: Half-moon dataset (ground truth)

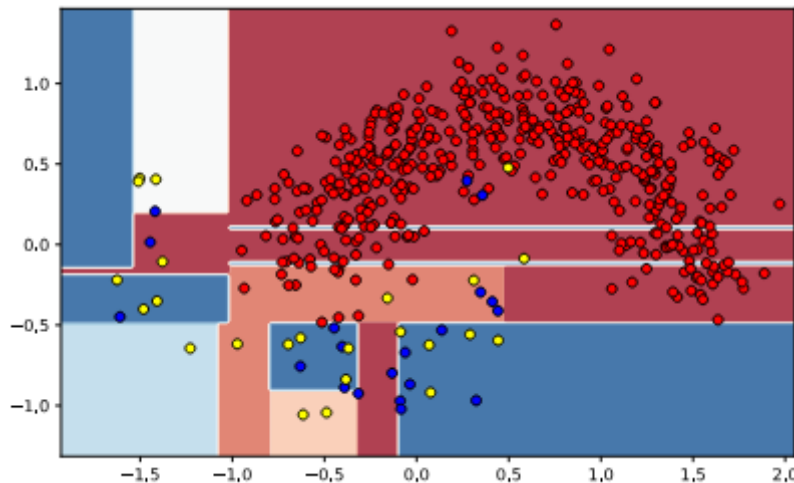
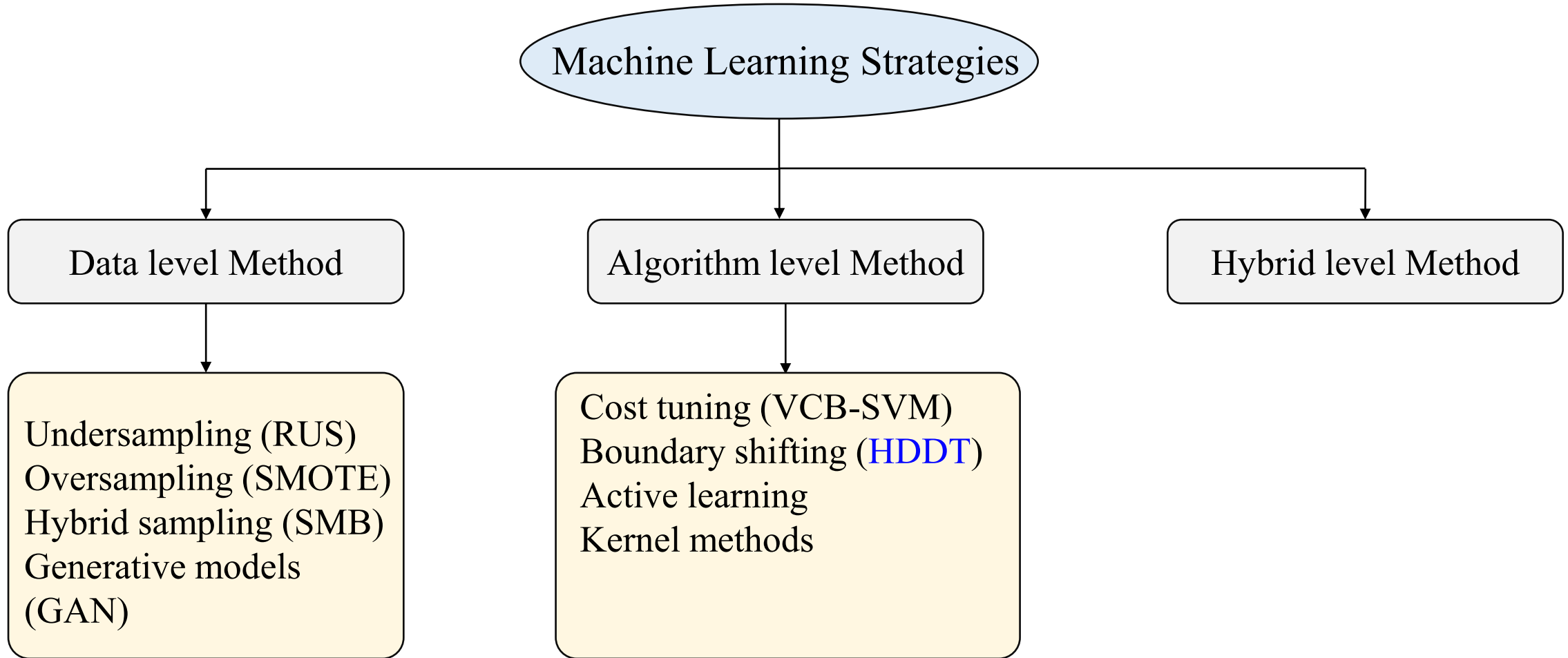


Fig: CART on test dataset

- Decision tree is biased towards the **majority** class due to the improper choice of splitting criteria for class imbalance problems.
- Standard splitting criteria, for e.g., IG and Gini Index, are **skew sensitive**.
- The red and orange dots represent the majority and hidden minority whereas the blue ones refer to the minority class. The areas with darker color (blue or red) points out more certainty regarding the classification into the **minority (blue)** or **majority (red)**. The lighter color, consequently, implies a higher uncertainty of the classifier.

Popular Class Imbalance Handling Techniques



Class Imbalance: Popular Handling Techniques

- **Over-sampling** - SMOTE (*Synthetic Minority Over-sampling Technique*)
 - Training complexity increases, the extent of SMOTE to be employed has to be tuned, can lead to **overgeneralization** of the minority class and can be problematic for highly skewed distribution of the classes. Recently, GAN is used as an alternative to SMOTE.
- **Under-sampling** - RUS (*Random Under-sampling*)
 - Critical points may be lost, the extent of RUS also has to be tuned.
- **Cost-sensitive Methods**
 - Can be computationally intensive.
- **Algorithm level approaches:** Kernel modification methods, active learning methods, rank metrics and multitask learning etc.
 - Mostly heuristic and suboptimal, sometimes no theoretical performance guarantee!
- **Hybrid level methods:** Combination of two or more of the above.

Hellinger Distance Decision Tree (HDDT)

- HD is symmetric, non-negative, and skew insensitive.
- Let (Θ, λ) denote a measurable space. Let us suppose that P and Q are two continuous distributions with respect to the parameter λ having the densities p and q in a continuous space Ω , respectively. Define HD as follows:

$$d_H(P, Q) = \sqrt{\int_{\Omega} (\sqrt{p} - \sqrt{q})^2 d\lambda}$$

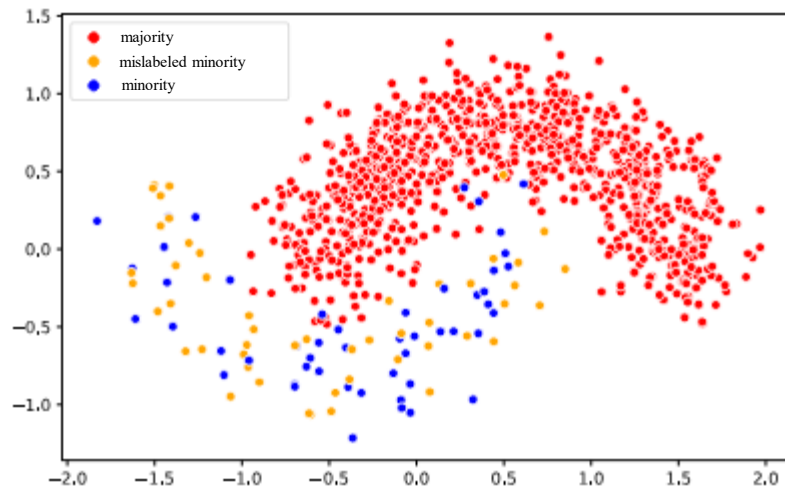
- Discrete case:

$$HD_i = \sqrt{\left\{ \sqrt{\frac{N_{left_i}}{N_i}} - \sqrt{\frac{P_{left_i}}{P_i}} \right\}^2 + \left\{ \sqrt{\frac{N_{right_i}}{N_i}} - \sqrt{\frac{P_{right_i}}{P_i}} \right\}^2},$$

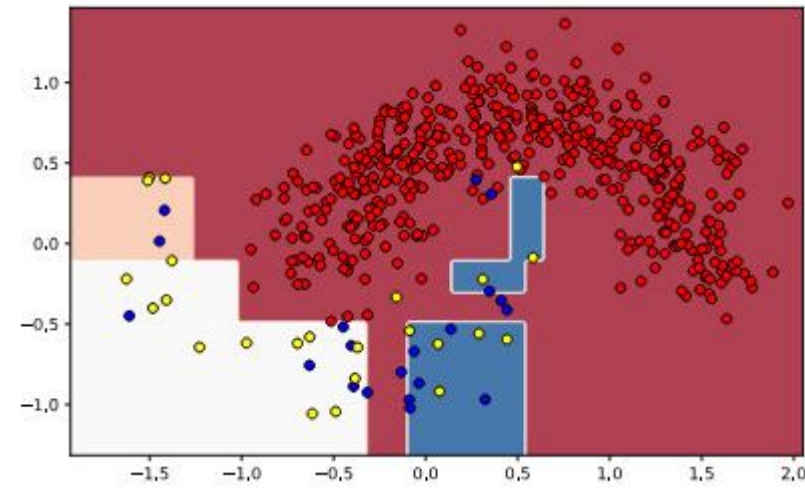
where N_i is the count of majority samples in the parent node i ;
 N_{left_i} is the count of majority samples that fall into the left child node.

- The aim is to split tree nodes on those features with minimal *affinity*, i.e., maximal HD.

Hellinger Distance Decision Tree (HDDT)



(a) Half-moon dataset (ground truth)



(b) HDDT on test dataset

Fig: Decision boundary of HDDT in an imbalanced classification problem.

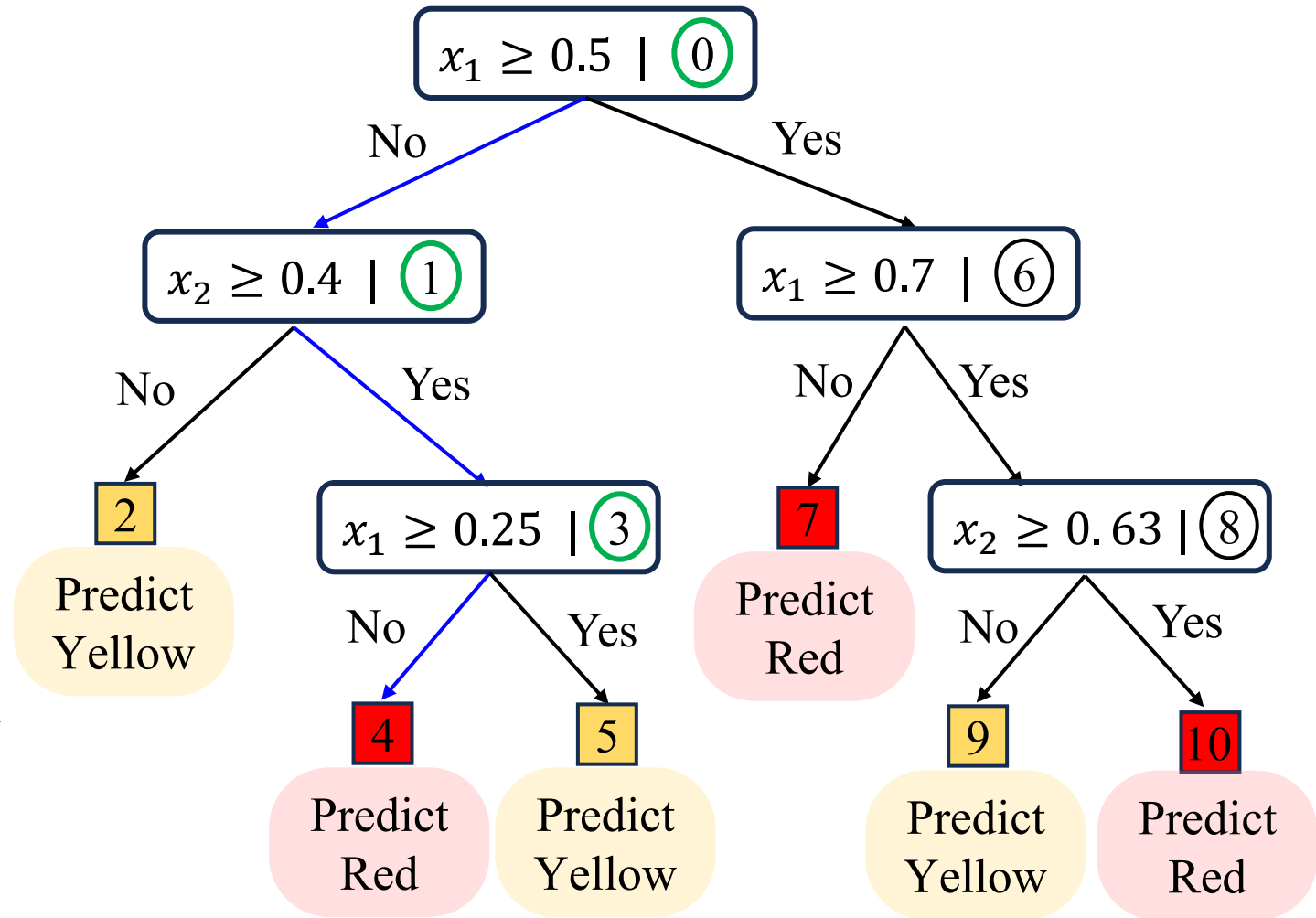
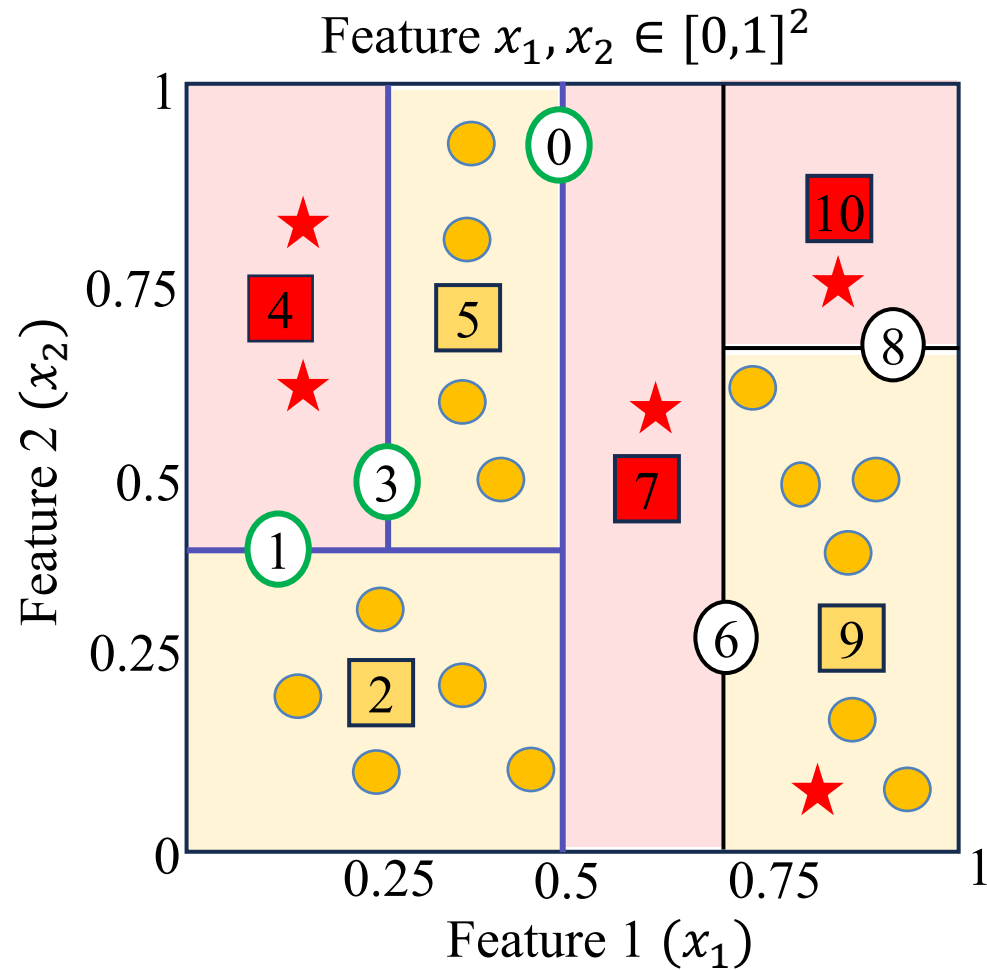
- **Key Drawbacks:**

1. HDDT builds unpruned tree; therefore, overfits.
2. During local search, it may get stuck in local optima.
3. Generalization issues and restricted decision boundaries.

Proposed Solution: Hellinger Net

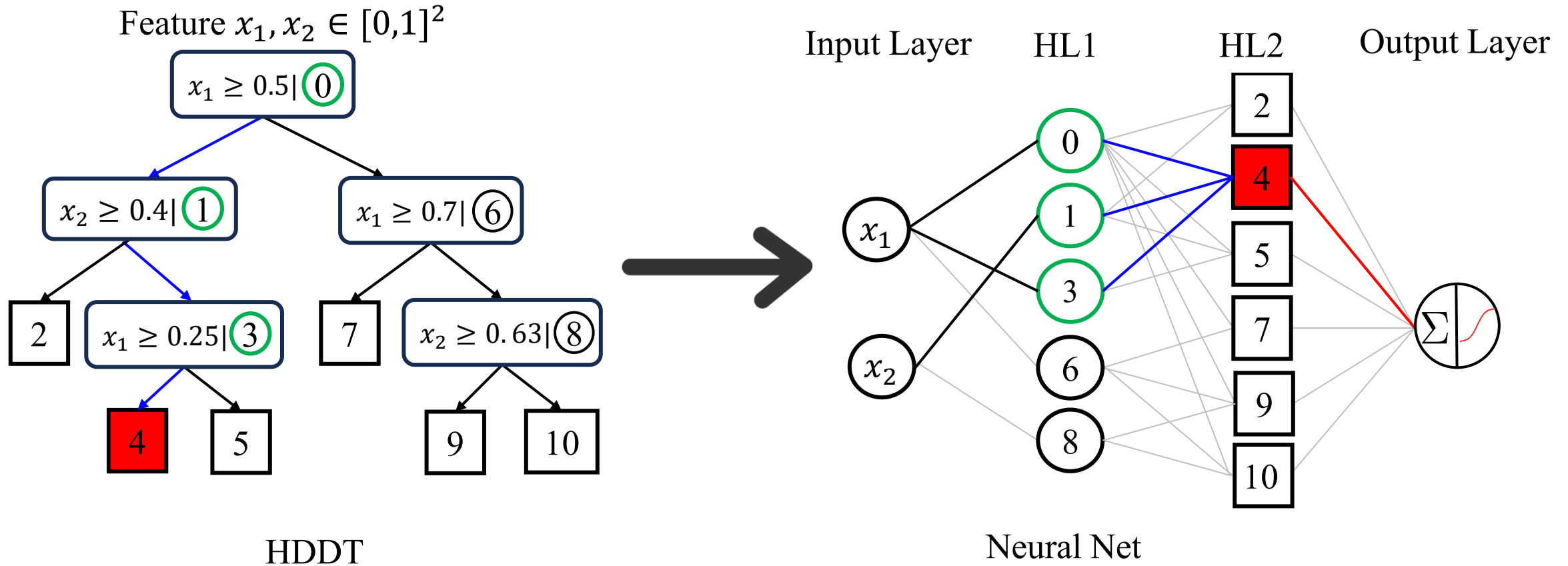
- We propose a new hybrid procedure to handle data imbalance that will work for limited data.
- Hellinger Net (**HNet**) is composed of three basic steps:
 - Step-1: Converting a DT into rules (**HD** is used as split criterion)
 - Step 2: Constructing two hidden layered NN from the rules
 - Step 3: Training the MLP using SGD backpropagation (less restrictions on the decision boundaries than trees)
- Instead of using pruning methods (removing child nodes), **HNet** employs a backpropagation NN to give weights to nodes according to their significance.
- HNet provides better **generalization** and **smooth decision boundaries** as opposed to HDDT.

Step1: Converting Tree into Rules



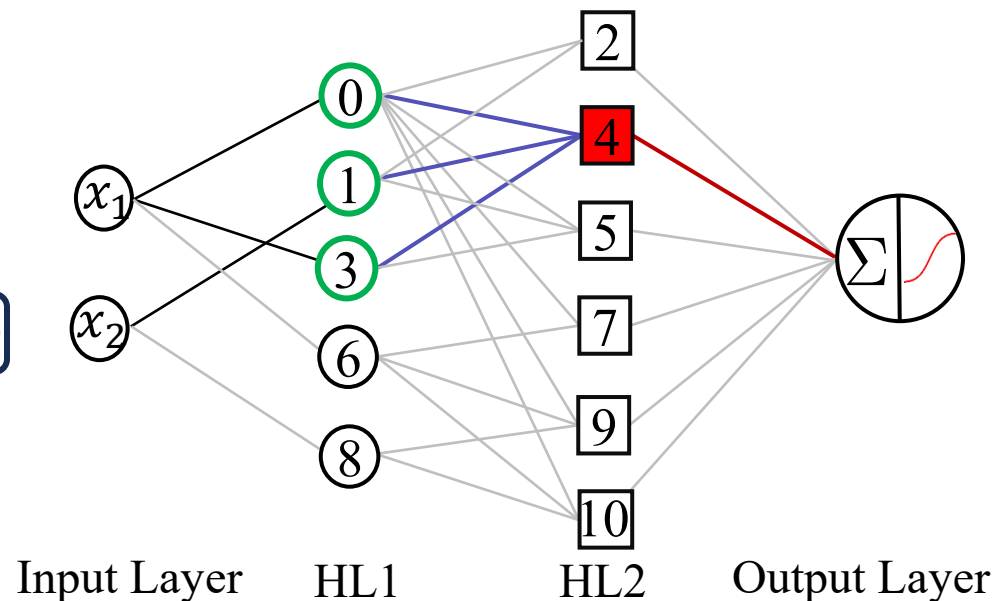
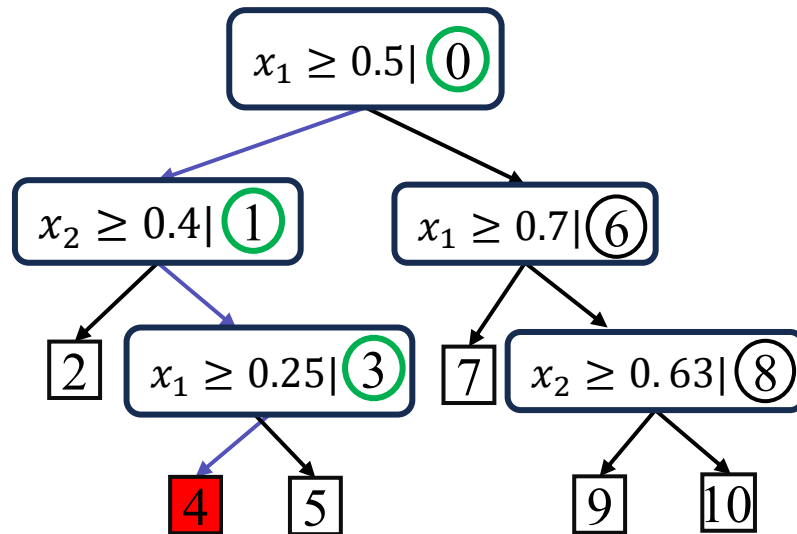
Step 2: From HDDT to Neural Net

- Converting a decision tree to a NN is to exploit **prior knowledge of HDDT** and fix the network architecture (have **fewer parameters to tune** than standard networks)



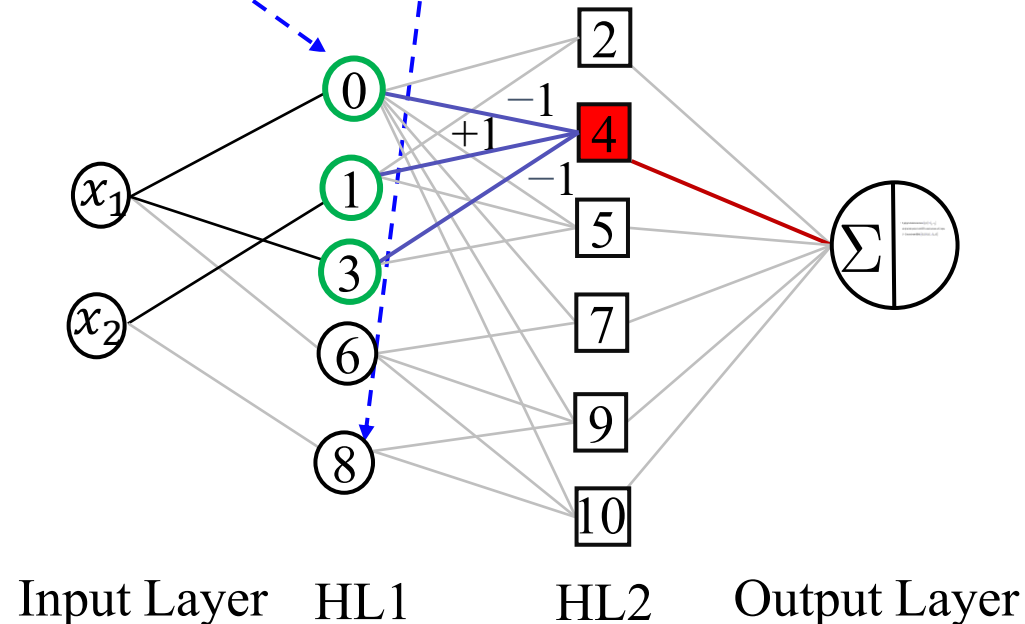
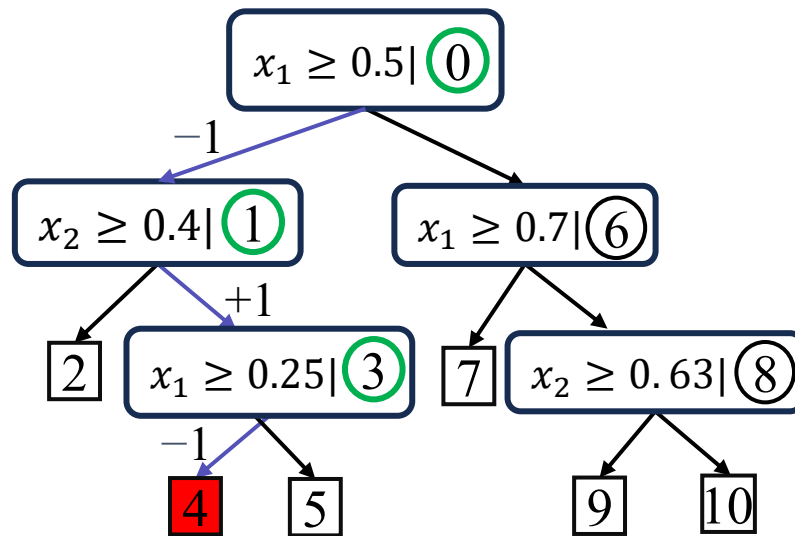
Step 2: Input & First Hidden Layer

- The input layer supplies the features (x_i) to HL1, which corresponds to $k - 1$ hidden neurons (since HDDT is a complete binary tree, it has $k - 1$ internal nodes, and k leaves)
- The first hidden layer (HL1 = $\{H_1, H_2, \dots, H_{k-1}\}$) consists of hyperplanes
 $H_{k'} = \{x \in [0,1]^p: h_{k'}(x) \geq 0\}$ with $h_{k'}(x) = x_{i_{k'}} - \alpha_{i_{k'}}$
- The threshold activation function is applied $\tau(z) = 2 \times I_{z \geq 0} - 1$



Step 2: First Hidden Layer output

- By applying the threshold activation function $\tau(h_{k'}(x)) = \tau(x_{i_{k'}} - \alpha_{i_{k'}})$ each split and relative position of $\mathbf{x}$ in the HDDT is encoded in each neuron, so HL1 outputs a $(k - 1)$ dimensional **vector of ± 1 bits**, $(\tau(h_1(x)), \tau(h_2(x)), \dots, \tau(h_{k-1}(x)))$.
- Each neuron of HL1 outputs $u_{k'}(x) = \tau(h_{k'}(x))$ takes the value +1 if x falls on the right side ($h(x) \geq 0$), and -1 otherwise.



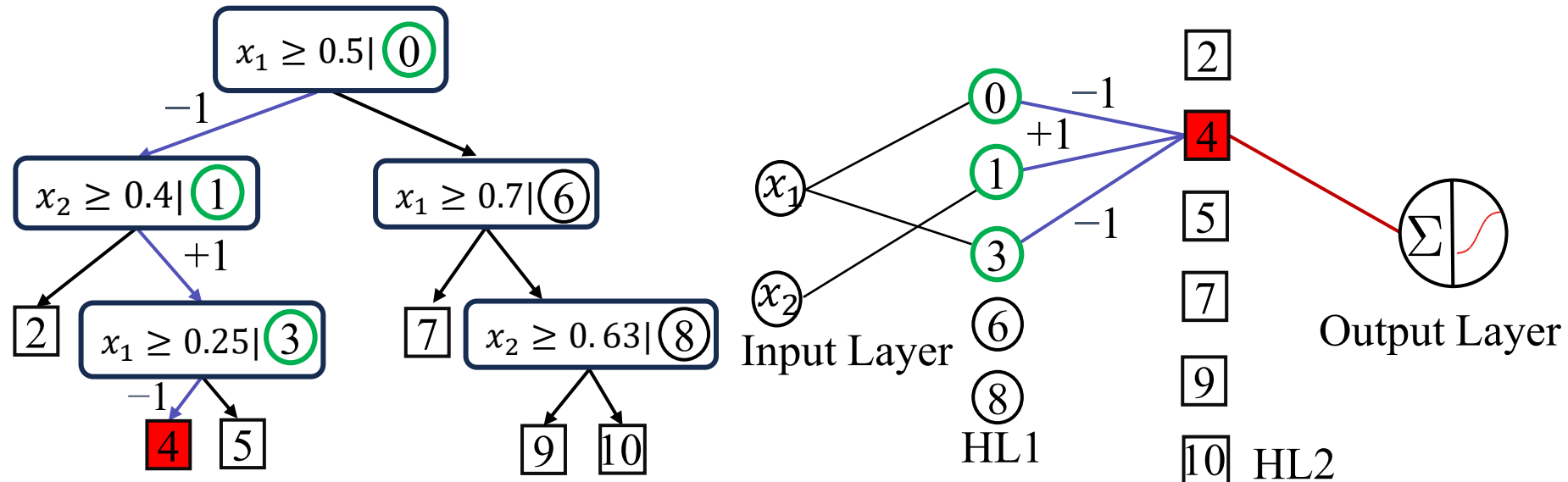
Step 2: Second Hidden Layer - ANDing

- The second hidden layer (HL2 = $\{L_1, L_2, \dots, L_k\}$) comprises k leaves, and each connection from k' of HL1 to k'' of HL2.
- The outputs of HL2 is a k dimensional **vector of ± 1 bits**, $(v_1(\mathbf{x}), v_2(\mathbf{x}), \dots, v_k(\mathbf{x}))$ where

$$v_{k''}(x) = \tau \left(\sum_{k' \rightarrow k''} b_{k',k''} u_{k'}(x) + b_{k''}^0 \right),$$

where $k' \rightarrow k''$ mean k' is connected to k'' , $b_{k',k''} = \pm 1$ is the corresponding weight, and $b_{k''}^0$ is the offset.

- $v_{k''}(x) = 1$ if and only if the terminal cell of the value x is $L_{k''}$.



Step 2: Output Layer - “one-hot” vector OR-ing

- The output layer takes the majority vote of the class labels falling in $L_{k''}$ when $v_{k''}(x) = +1$.
- All activities of HL2 are linearly rescaled from $[-1, 1]$ to $[0, 1]$.
- The weights from HL2 to the output layer are set proportional to the majority votes.

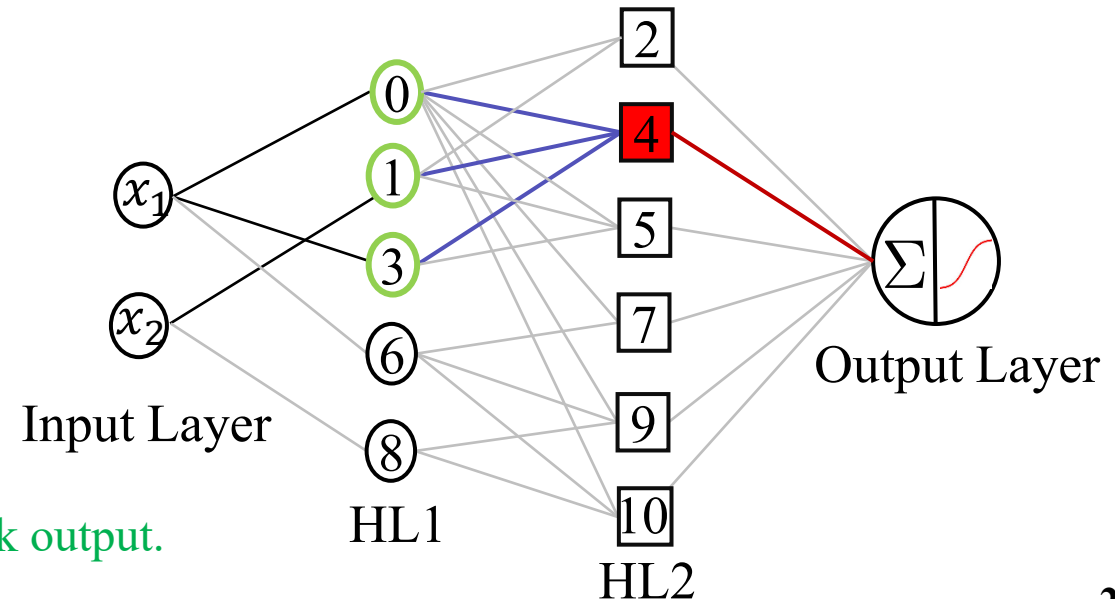
- Tree estimate:
$$t_n(x) = \sum_{k''=1}^k w_{k''} v_{k''}(x) + b_{\text{out}}, \quad \text{with } w_{k''} = \frac{\bar{Y}_{k''}}{2}, \quad b_{\text{out}} = \frac{1}{2} \sum_{k''=1}^k \bar{Y}_{k''}$$

$$= \bar{Y}_{k''}, \quad \text{when } v_{k''}(x) = 1.$$

- Classification rule using soft-max activation function:

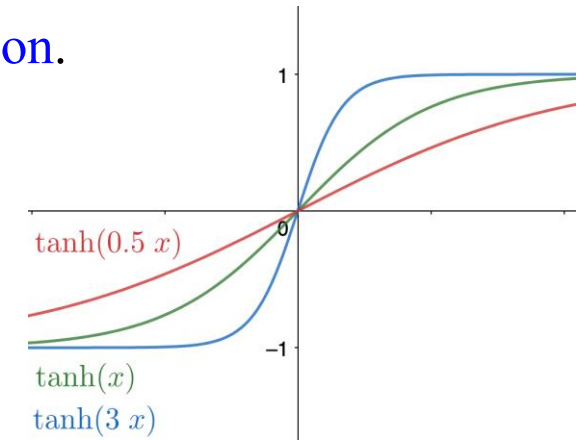
$$g_n(x) = \begin{cases} 0, & \text{if } t_n(x) \leq \frac{1}{2} \\ 1, & \text{otherwise} \end{cases}$$

- The predicted class is $g_n(x)$ (majority or minority).
- HNet provides a probabilistic interpretation of the neural network output.



Step 3: Neural Network Training

- The HNet uses **tanh** activation functions in HL1 and HL2 instead of threshold ones (the loss function becomes differentiable everywhere): $\tau(z)$ will be replaced by $\sigma(z) := \tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$
- HL1 uses $\sigma_1 = \sigma(\beta_1 u)$, while HL2 uses $\sigma_2 = \sigma(\beta_2 u)$ for better generalization.
- β_1 and β_2 are positive hyperparameters that determine the contrast of the hyperbolic tanh activation function.
- The larger β_1, β_2 , the sharper the transition from -1 to +1.
- When β_1, β_2 approaches to ∞ , σ_1, σ_2 converges to the threshold function.
- The Hellinger net is trained using **stochastic gradient descent backpropagation** algorithm.
- Since the number of neurons in the hidden layers is fixed, thus the training time is less.



Merits



Feature Algorithm	High- dimensional setting	Non-restrictive Decision Boundary	Restrict Overfitting	High Accuracy	Interpret- ability	Imbalance Handling
Decision Tree	✓	✗	✗	✗	✓	✗
Hellinger Distance DT	✓	✗	✗	✓	✓	✓
Neural Net	✗	✓	✗	✗	✗	✗
Hellinger Net (Proposed)	✓	✓	✓	✓	✓	✓

On Theoretical Consistency



Theorem (Chakraborty et al., 2021, IEEE Trans. on Reliability)

Assume X is uniformly distributed in $[0,1]^p$ and $Y = \{0,1\}$. As $n \rightarrow \infty$ and for any $k_n, \beta_1, \beta_2 \rightarrow \infty$ if the following conditions are satisfied:

$$(A1) \frac{k_n^4 \log(\beta_2 k_n^4)}{n} \rightarrow 0,$$

$$(A2) \text{ there exists } \delta > 0 \text{ such that } \frac{k_n^2}{n^{1-\delta}} \rightarrow 0,$$

$$(A3) \frac{k_n^2}{e^{2\beta_2}} \rightarrow 0, \text{ and}$$

$$(A4) \frac{k_n^3 \beta_2}{\beta_1} \rightarrow 0,$$

then *Hellinger Nets* classifier is consistent.

The above Theorem states that with certain restrictions imposed on the number k_n of terminal nodes and the parameters β_1, β_2 being properly regulated as functions of n , the empirical L_1 risk-minimization provides local consistency of the Hellinger Nets classifier. The larger the values of k_n, β_1, β_2 the better is the model in practice.

Results on Simulated Dataset



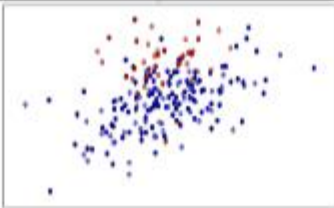
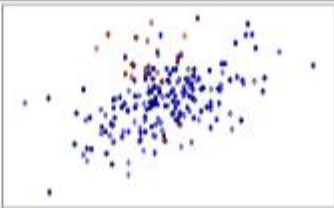
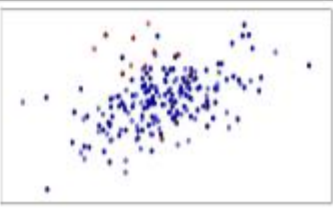
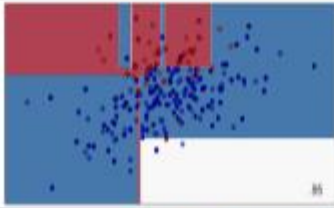
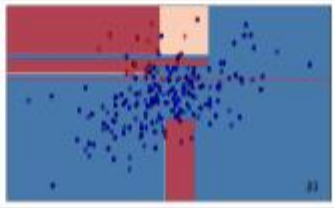
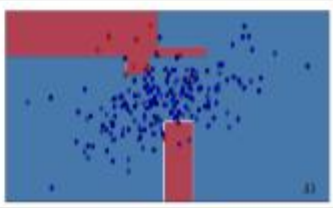
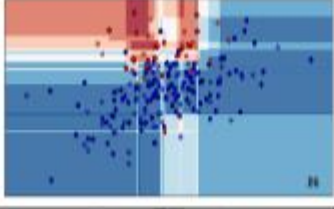
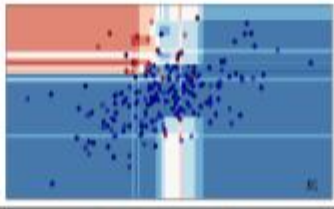
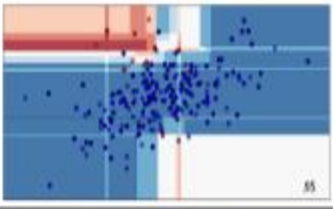
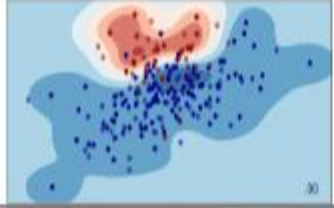
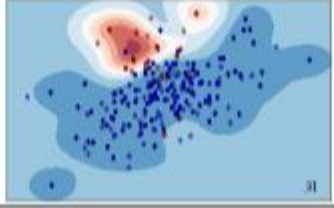
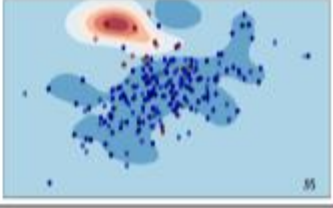
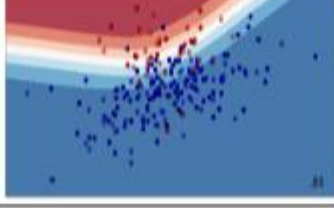
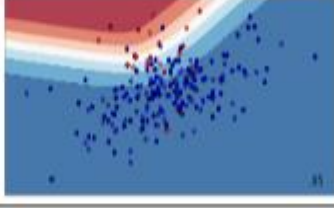
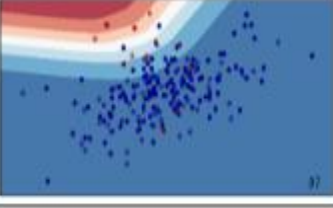
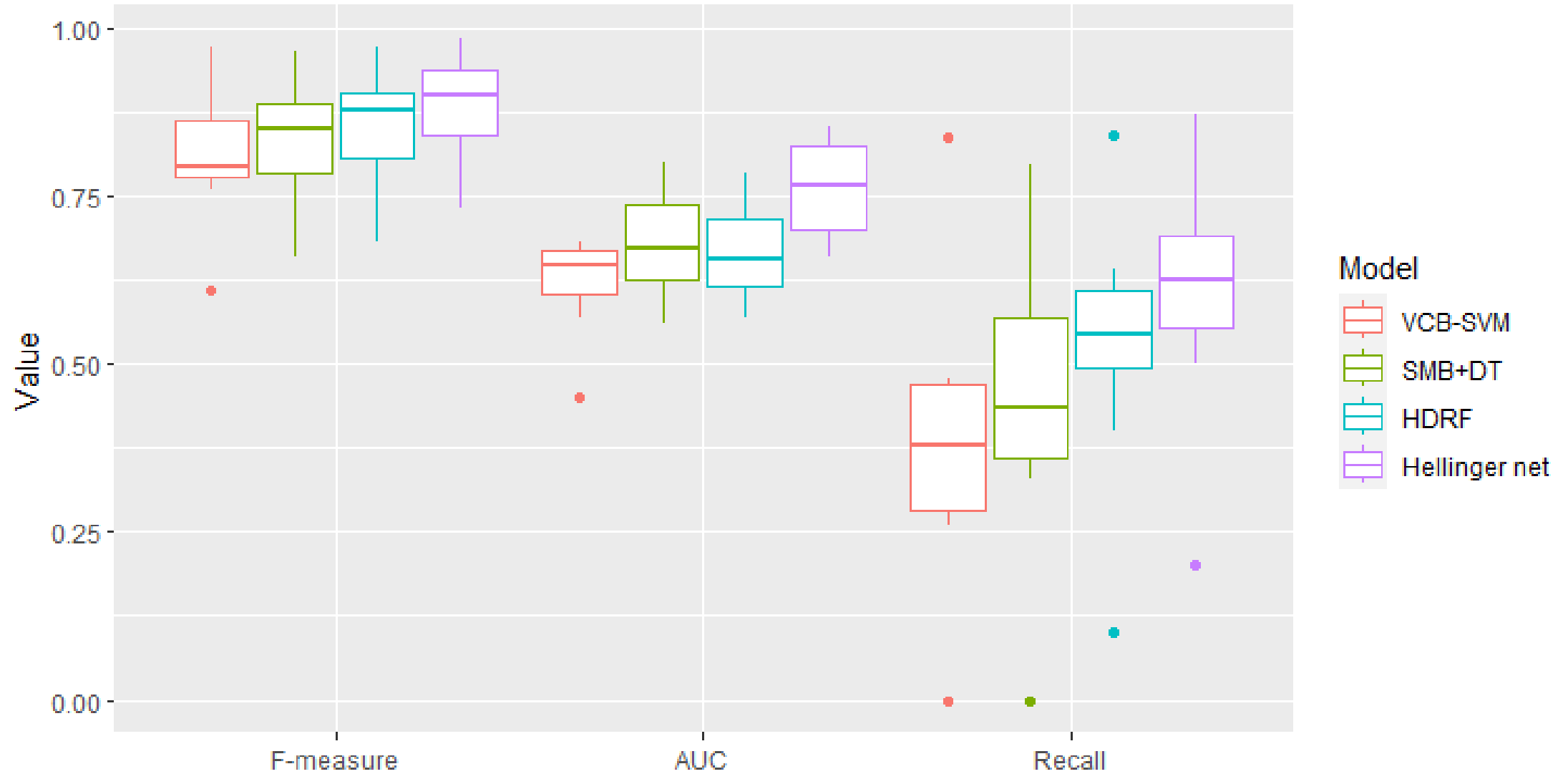
Synthetic data	Imbalance = 20% rate	Imbalance = 10% rate	Imbalance = 5% rate
Input Data			
HDDT			
HDRF			
VCB-SVM			
Hellinger net			

Table: A comparison of several imbalanced classifiers on synthetic data sets. The plots show training points in solid colors and testing points semi-transparent. The lower right in each plots shows the AUC value on the test set.

- AUC estimates the area under the ROC curve that illustrates the trade-off between TPR and FPR.

Results on NASA SDP Datasets



Results on Thyroid Gland Dataset

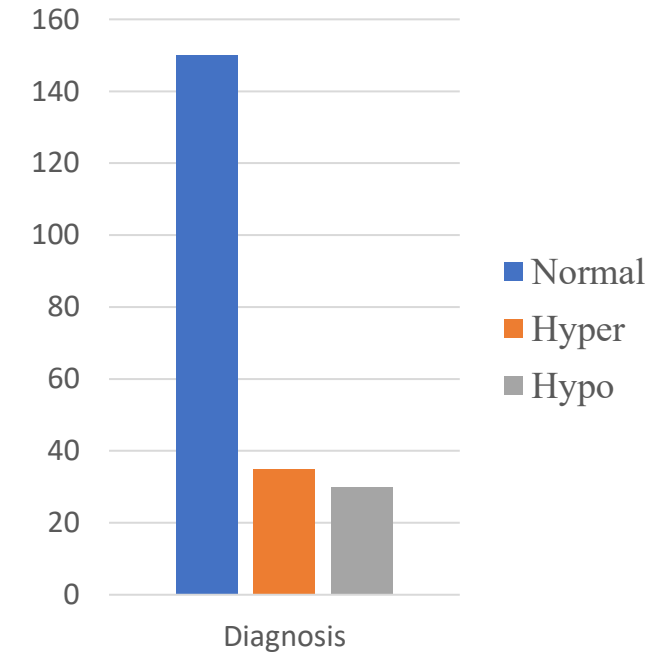
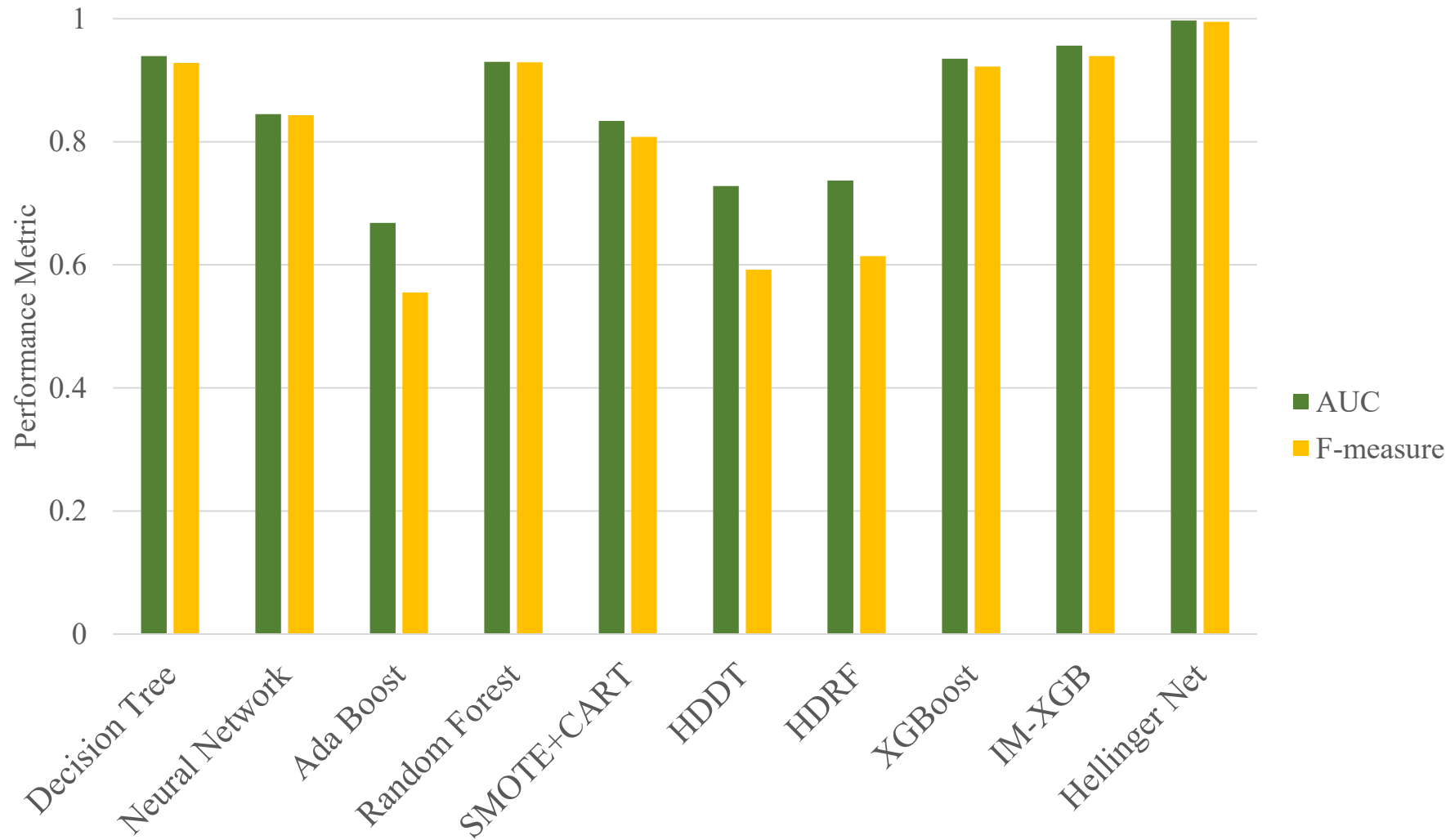


Fig: Output class distribution

Remarks



This work proposed ‘Hellinger Net’, a hybrid learner, that first constructs a tree and then simulates it using neural networks to solve imbalanced learning problem with limited datasets. Python Code on [GitHub](#).

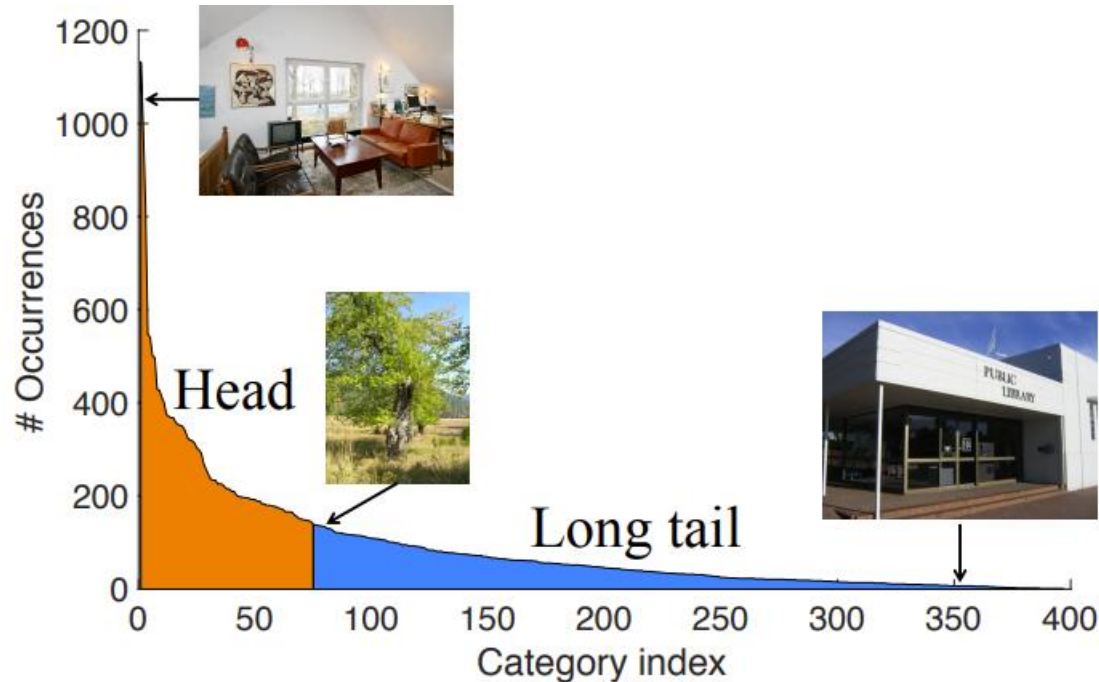
- **Limitations:**

1. It is not straight forward to extend this to more than two layers.
2. The algorithm may not work well for **HDLSS ($p > n$) imbalanced data problem** (e.g., medical image analysis and microarray gene expression studies).

Possible Solution: Regularization in Hellinger Net classifier.

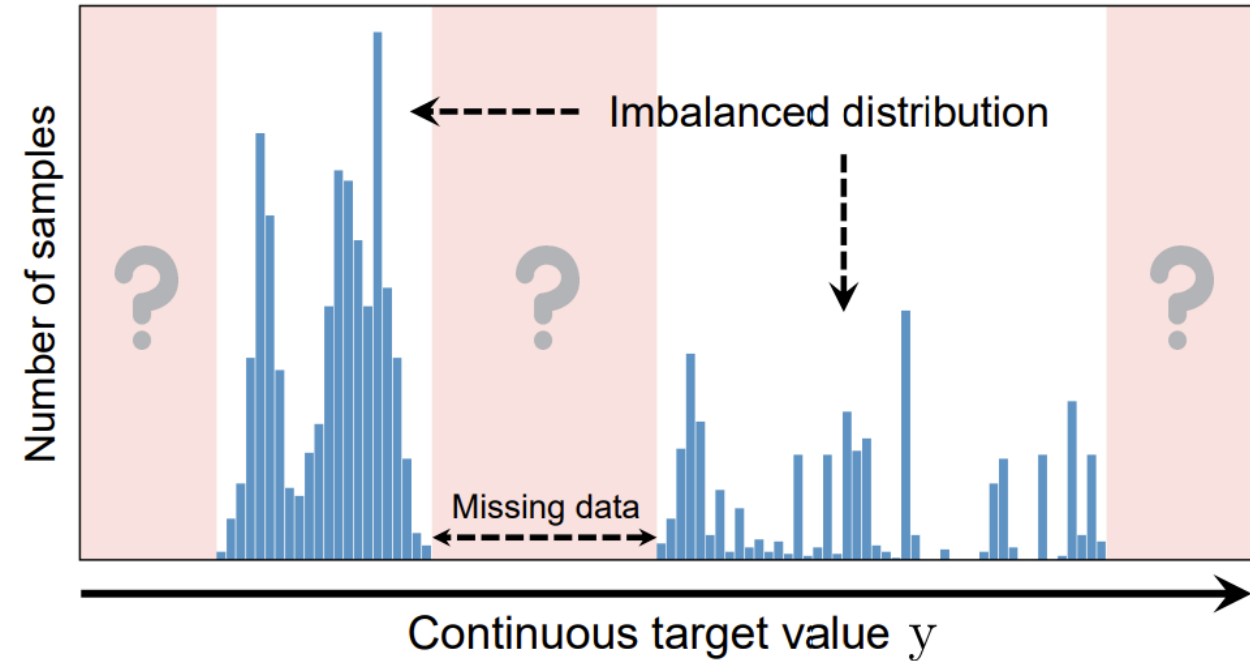
- **Business Impact:** The method was implemented for **ISRO SDP** project and a modified version of HNet has been adopted by **Bajaj Finserv Pvt. Ltd.** for identifying loan defaulters.

Imbalanced Learning: What lies ahead?



(a) Long-tail distribution on the SUN-397 dataset.

Ref: Wang et al. (2017) "Learning to model the Tail."
NIPS.



(b) Imbalanced data with continuous targets

Ref: Yang et al. (2021) "Delving into Deep Imbalanced Regression." ICML.

Key Papers on Imbalanced Learning



IEEE TRANSACTIONS ON
RELIABILITY

published by the
IEEE RELIABILITY SOCIETY

Hellinger Net: A Hybrid Imbalance Learning Model to Improve Software Defect Prediction

Tanujit Chakraborty  and Ashis Kumar Chakraborty 



Contents lists available at [ScienceDirect](#)

Pattern Recognition

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Skew-probabilistic neural networks for learning from imbalanced data

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essentially,
all models are wrong,
but some are useful

George E. P. Box

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